

Solution Requirements

Date: 30 September 2026 **Team ID:** 6ab3e57da238dc999eb3d44d **Project Name:** EduGenie – AI-Powered Educational Assistant

Maximum Marks: 4 Marks

Functional Requirements

S.No	Category	Requirement Description	Priority
1	Authentication	No user authentication is implemented in the current version; the local application is accessible through the browser.	Low
2	Authorization	No role-based access is required for the current student-focused prototype.	Low
3	External interfaces	The backend connects to Google Gemini through the Google Gen AI SDK when a valid API key is configured.	High
4	Transactions processing	User text/topic is validated, routed to the selected module, processed by the AI service, and returned to the browser.	High
5	Reporting	The current release returns learning outputs directly; persistent analytics dashboards are future scope.	Medium
6	Business rules	Quiz generation must return three questions with exactly four options each; input lengths are validated.	High
7	Compliance	API keys must remain in environment variables; users are advised not to submit sensitive/private content.	High
8	Other	Five workflows: Q&A, Explain, Summarize, Quiz, and Learning Path recommendations.	High

Non-Functional Requirements

S.No	NFR	Requirement	Acceptance Criteria
1	Performance & Speed	Keep the web interface responsive and avoid unnecessary backend work.	Successful local requests return without critical server errors.
2	Scalability	Use modular feature modules and a shared AI service so new workflows can be added independently.	A new module can be connected through a focused route without redesigning the whole application.
3	Security & Privacy	Store GEMINI_API_KEY outside source code and validate request sizes.	No API key committed to source; invalid/oversized inputs are rejected.
4	Reliability	Provide demo fallback when Gemini is unavailable and useful errors for invalid quiz output.	Application can demonstrate core UI flow without a live API key.
5	Usability	Provide one browser interface with clear actions and result display.	User can select a workflow, submit input and see a result.
6	Maintainability	Separate API routing, AI provider logic and learning modules.	Responsibilities are separated across main.py, ai_service.py and feature modules.